AWS GLUE ETL PIPELINE WITH AMAZON S3

Project Overview

This project demonstrates an Extract, Transform, and Load (ETL) pipeline using AWS Glue and Amazon S3. Raw CSV data is stored in an S3 bucket, discovered using an AWS Glue Crawler, transformed using an AWS Glue ETL job, and written back to S3 as curated data.

Technology Used

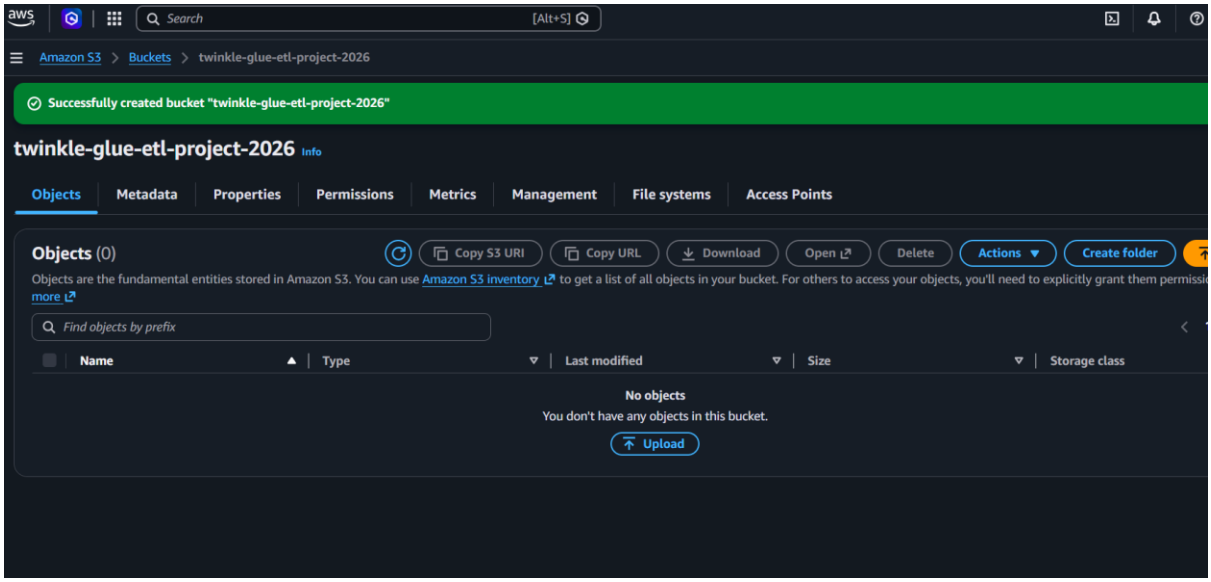
- Amazon Web Services (AWS)
- Amazon S3
- AWS Glue
- AWS Glue Data Catalog
- Python and PySpark
- CSV and Parquet

AWS Region: Asia Pacific (Mumbai) — ap-south-1

Prepared by: Twinkle Meska

1. Amazon S3 Bucket Creation

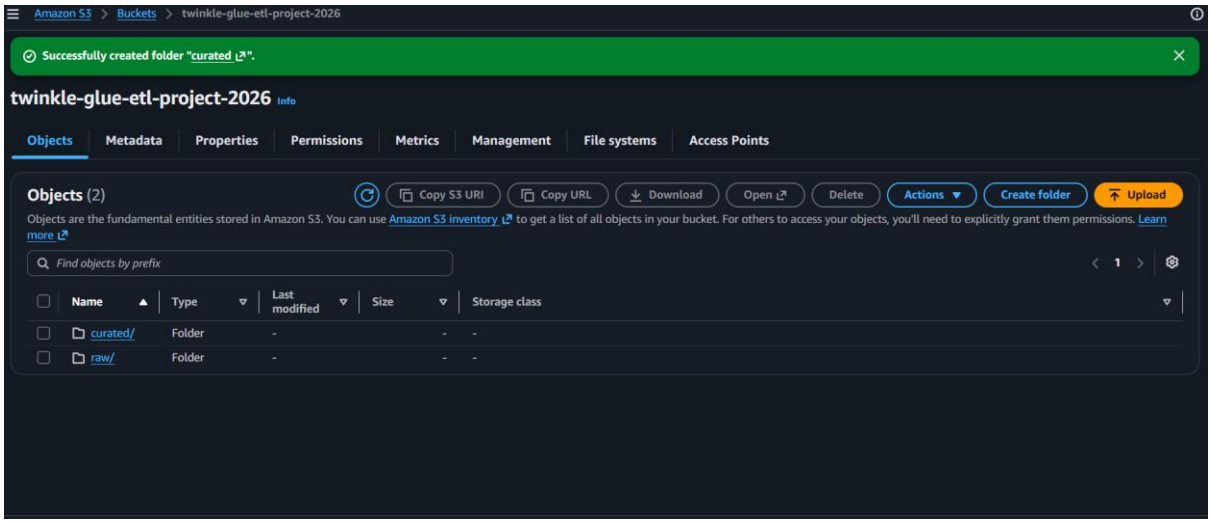
The first step in building the ETL pipeline is to create an Amazon S3 bucket to store raw input data and transformed output data. A dedicated bucket named twinkle-glue-etl-project-2026 is created for this project. Separate folders will be used to organize the raw and curated datasets.



2. Creating Raw and Curated Folders in Amazon S3

Two folders, raw/ and curated/, were created inside the Amazon S3 bucket to organize the ETL pipeline. The raw/ folder stores the original input dataset, while the curated/ folder is the destination for the transformed output generated by AWS Glue.

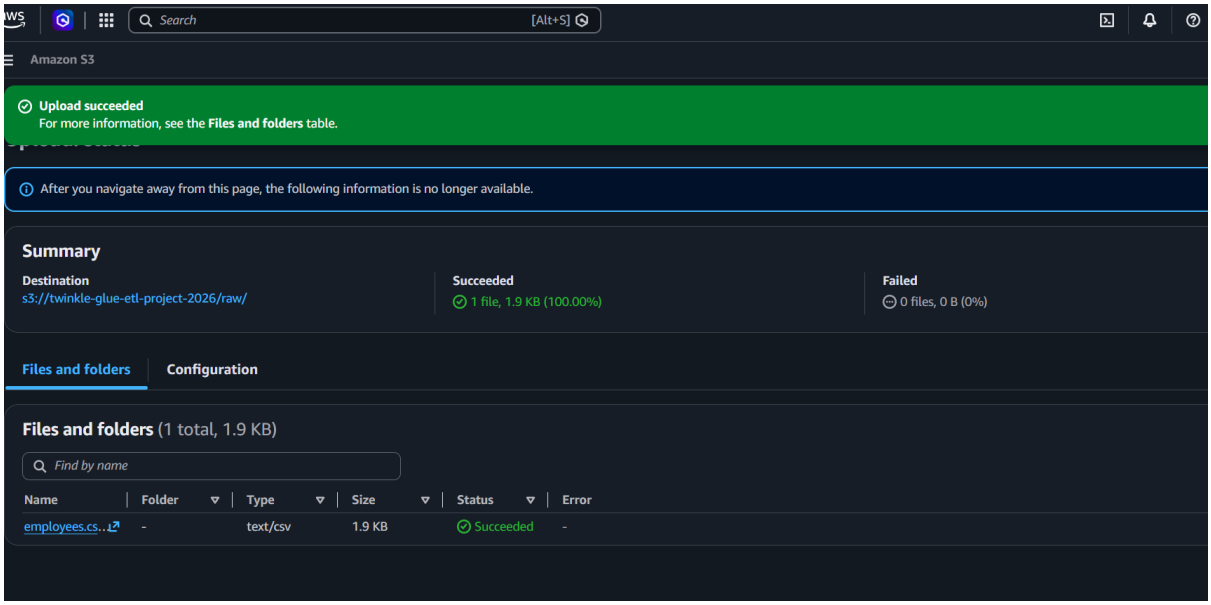
Raw and curated folders created in the Amazon S3 bucket.



3. Uploading the Employee Dataset to Amazon S3

A sample employee dataset named employees.csv was uploaded to the raw/ folder in the Amazon S3 bucket. The dataset contains employee IDs, names, departments, salaries, and cities. This file serves as the source dataset for the AWS Glue ETL pipeline.

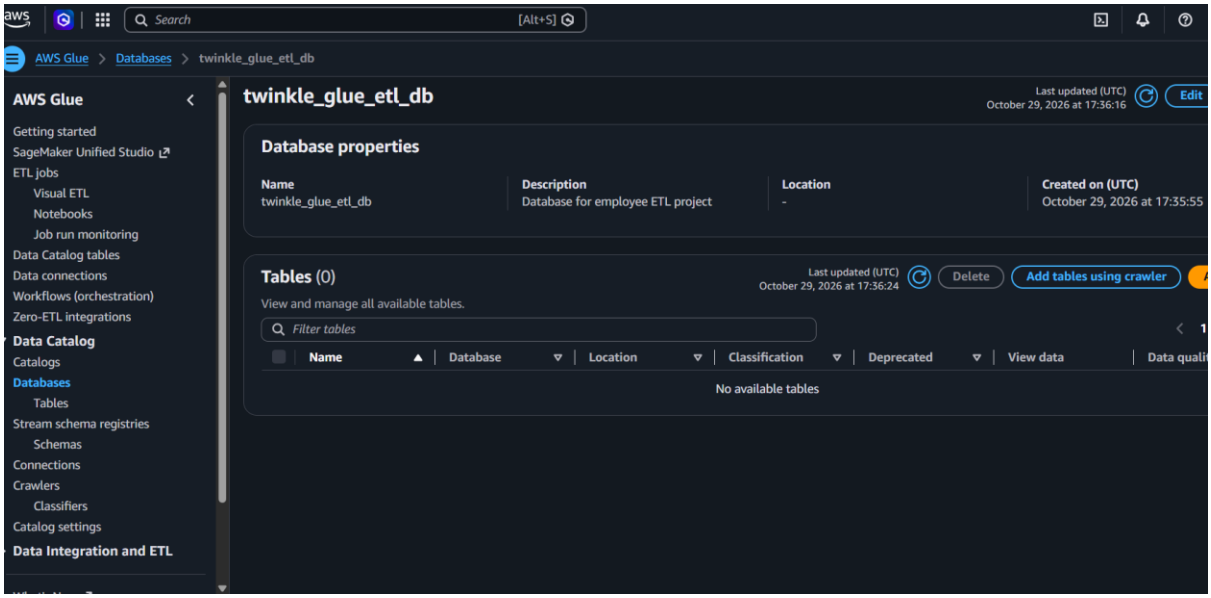
Employee CSV dataset uploaded to the raw folder in Amazon S3.



4. Creating an AWS Glue Data Catalog Database

An AWS Glue Data Catalog database named twinkle_glue_etl_db was created to organize the metadata of the employee dataset. The database will contain the table created by the AWS Glue Crawler after it examines the CSV file stored in Amazon S3.

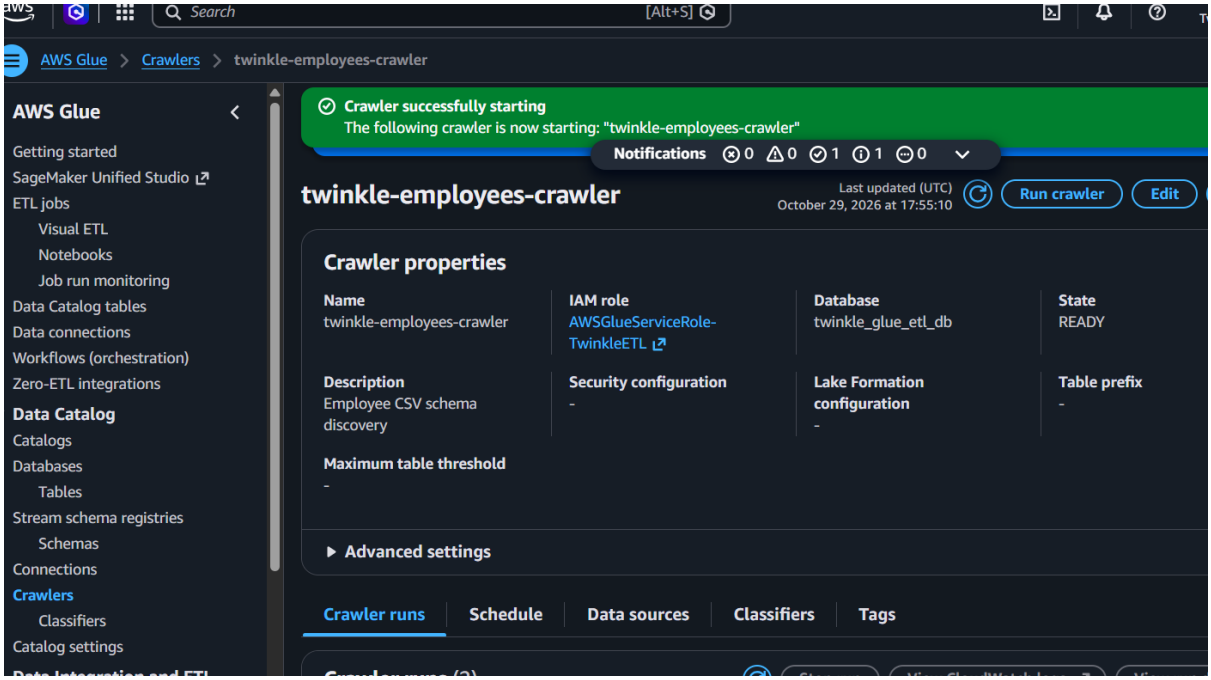
AWS Glue Data Catalog database created successfully.

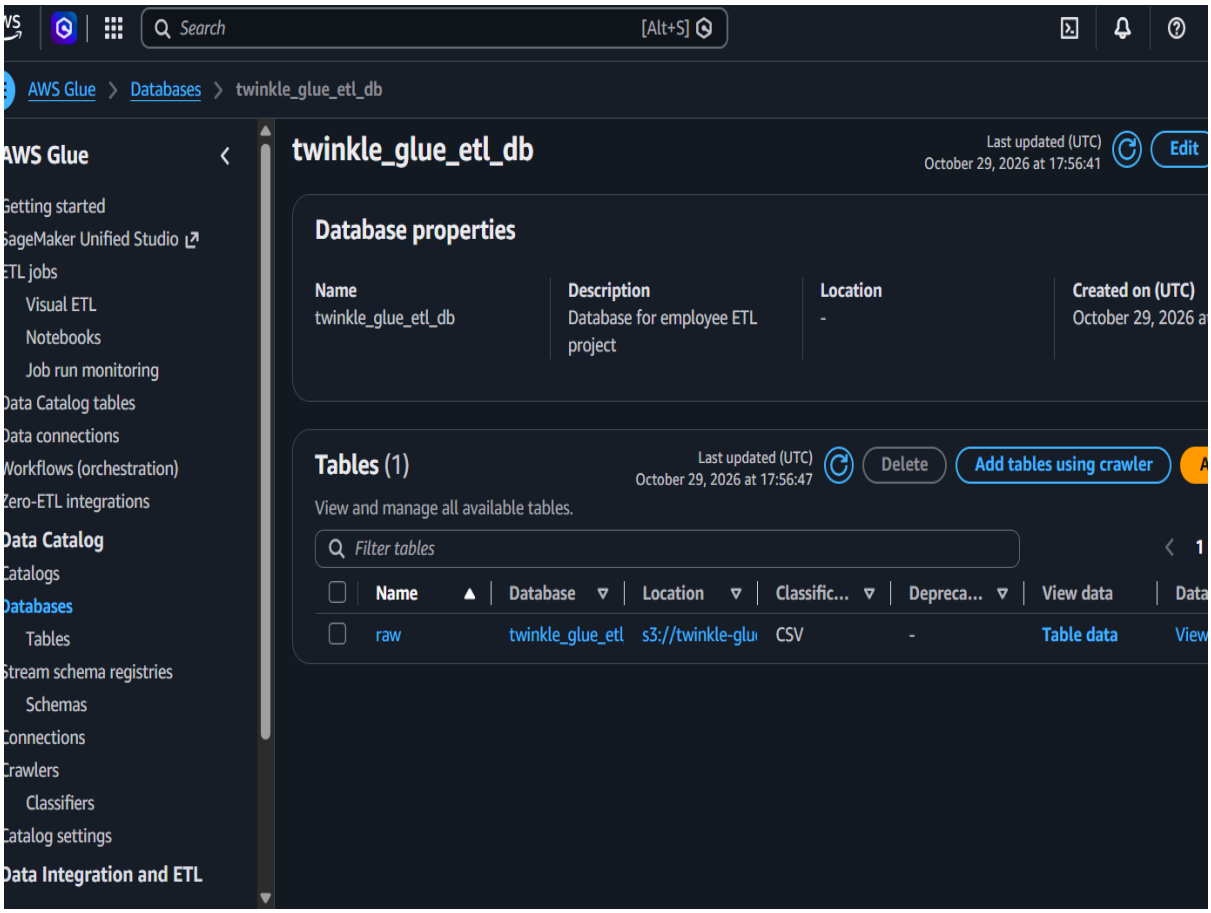


5. AWS Glue Crawler Execution and Schema Discovery

The AWS Glue Crawler was configured to scan the employee CSV dataset stored in Amazon S3. After a successful crawl, AWS Glue Data Catalog stores the discovered table metadata, including the column names and data types. This metadata can be used by the ETL job to process the source dataset.

Successful AWS Glue Crawler execution and discovery of the employee dataset.

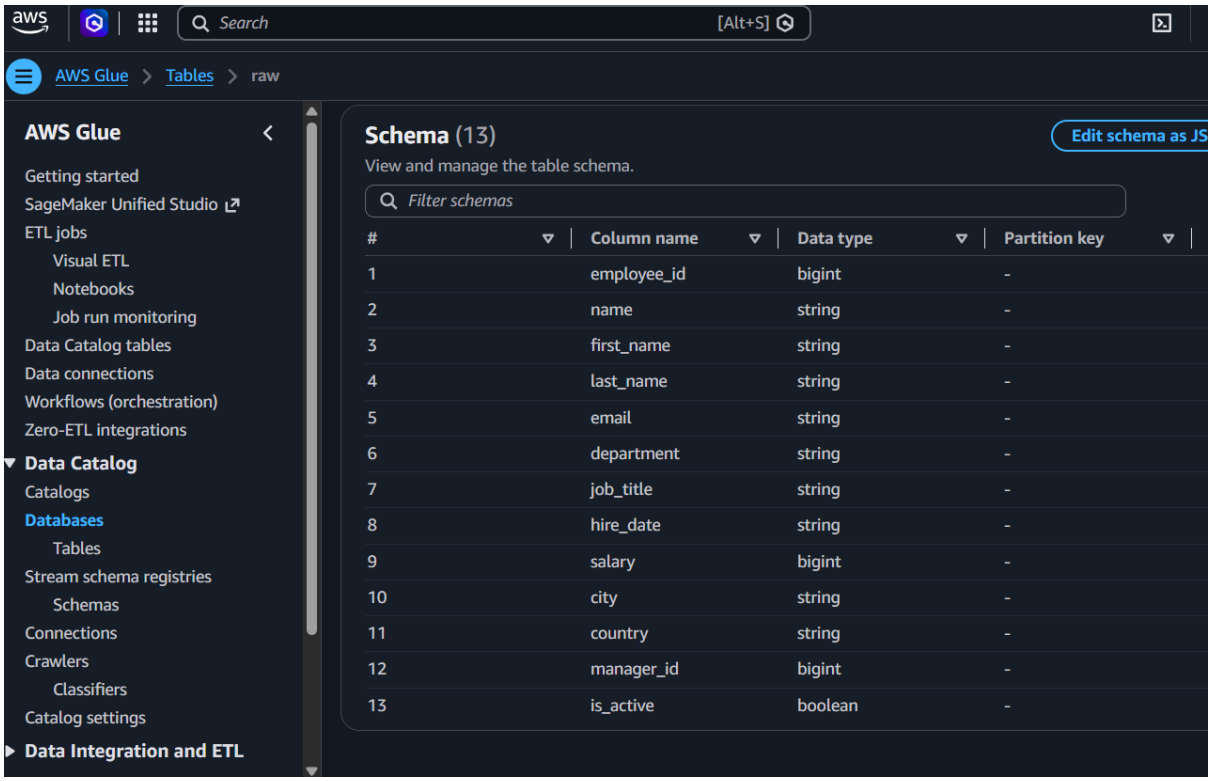

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Crawler successfully starting
The following crawler is now starting: "twinkle-employees-crawler"
Notifications 0 0 1 1 0
twinkle-employees-crawler
Last updated (UTC)
October 29, 2026 at 17:55:10
Run crawler
Edit
Crawler properties
Name
twinkle-employees-crawler
IAM role
AWSGlueServiceRole-TwinkleETL
Database
twinkle_glue_etl_db
State
READY
Description
Employee CSV schema discovery
Security configuration
-
Lake Formation configuration
-
Table prefix
-
Maximum table threshold
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Data Integration and ETL
twinkle_glue_etl_db
Last updated (UTC)
October 29, 2026 at 17:56:41
Edit
Database properties
Name
twinkle_glue_etl_db
Description
Database for employee ETL project
Location
-
Created on (UTC)
October 29, 2026
Tables (1)
Last updated (UTC)
October 29, 2026 at 17:56:47
Delete
Add tables using crawler
View and manage all available tables.
Filter tables
Name
Database
Location
Classific...
Deprecа...
View data
raw
twinkle_glue_etl
s3://twinkle-glu
CSV
-
Table data

6. AWS Glue Data Catalog Table and Schema

After the crawler execution, the discovered employee dataset is registered as a table in the AWS Glue Data Catalog. The table metadata describes the structure of the CSV file, including employee ID, name, department, salary, and city. This metadata enables AWS Glue to reference the source data during ETL processing.

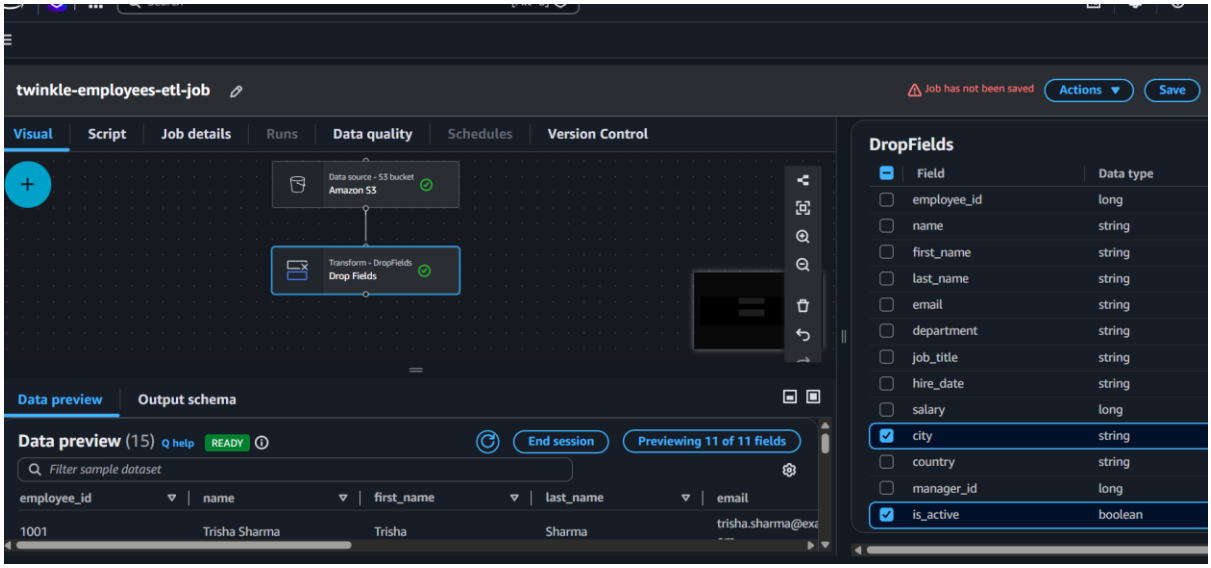
Employee dataset table and schema in the AWS Glue Data Catalog.



7. Configuring the Drop Fields Transformation

The Drop Fields transformation was added to the AWS Glue visual ETL job to remove the city column from the employee dataset while retaining the employee ID, name, department, and salary fields.

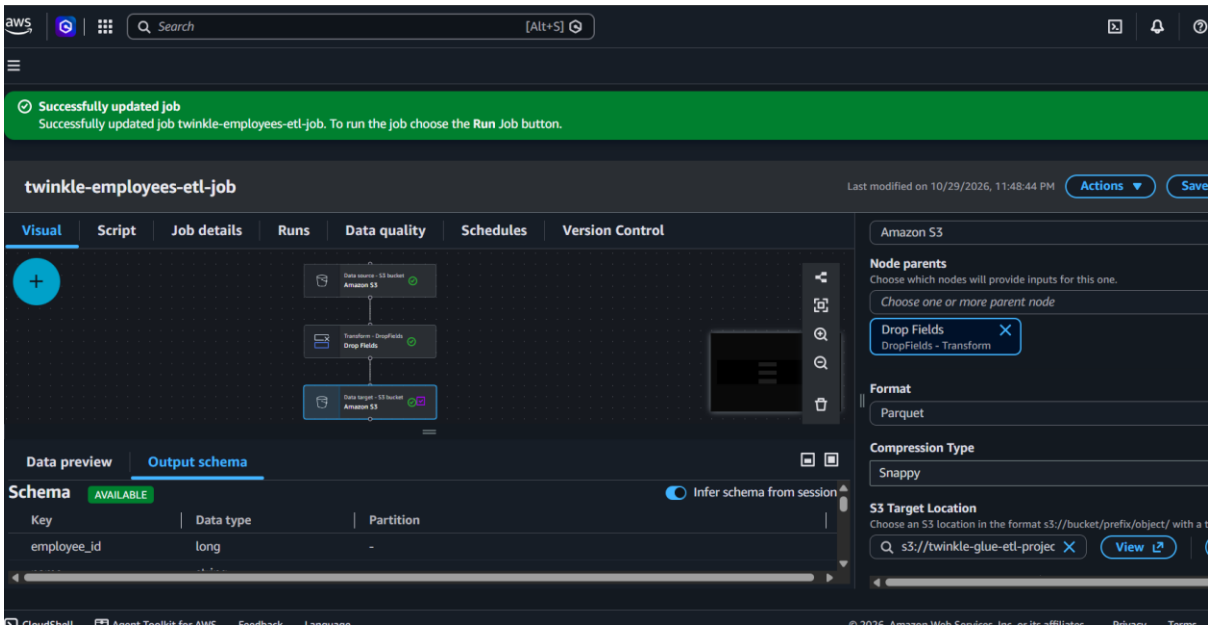
Drop Fields transformation configured to remove the city column.



8. Configuring the Amazon S3 Output Target

An Amazon S3 target was configured to store the transformed employee dataset in the curated/ folder of the project bucket. The output format was configured as Parquet for efficient storage and further data processing.

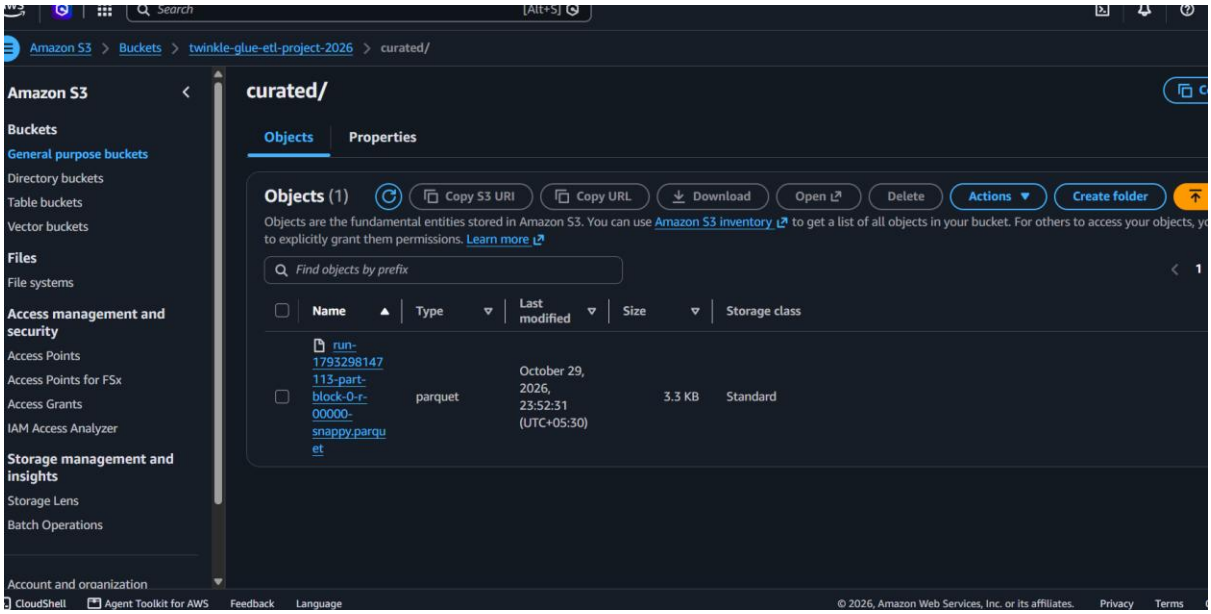
Amazon S3 target configured for the curated dataset in Parquet format.



9. Verification of Transformed Data in Amazon S3

After the AWS Glue ETL job executed, the transformed employee dataset was stored in the curated/ folder of the Amazon S3 bucket. The output file was generated in Parquet format with Snappy compression, demonstrating that the ETL job wrote the processed data to the target location.

Parquet output file generated in the curated folder of Amazon S3.



10. Successful Execution of the AWS Glue ETL Job

The AWS Glue ETL job was executed to process the employee dataset and write the transformed output to Amazon S3. The job run status was checked in AWS Glue Studio to verify whether the pipeline completed successfully.

AWS Glue ETL job run status and execution details.

